Machine Learning in Medicine Final Report

Machine-Learning Driven Identification of Pathogenic Predictors in Murine Cholelithiasis: A Comparative Study of Lithogenic Diet Response

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1. Project Abstract

Murine models of cholelithiasis provide a controlled setting to identify dietary and metabolic determinants of gallstone formation. We applied a machine learning pipeline to a multi-source phenotypic dataset comprising 385 obesity-prone mice to predict binary gallstone presence (*st_pres*) from morphometric, lipid, and strain-level summary variables. Individual phenotypes were synthesised with population-level strain summaries to provide a multi-scalar view of cholelithiasis risk. After a strict exclusion criterion for unlabeled instances ($n = 94$), we retained 291 animals with verified outcomes and implemented a multi-stage data refinement protocol. Three biologically motivated derived features—*weight_gain_pct*, *cholesterol_ratio*, and *non_hdl_calc*—were computed; leakage variables were excluded; columns with $> 70\%$ missingness were dropped. We compared Logistic Regression, Random Forest, and XGBoost under stratified 5-fold cross-validation, with recall prioritised to minimise false negatives. The champion model was a **Calibrated Logistic Regression** (recall 0.771, precision 0.615, F1 0.682), post-processed with isotonic calibration to ensure that predicted probabilities reflect observed event rates. SHAP analysis revealed that *totalCh* and *weight_gain_pct* are among the principal drivers of predicted risk. A central finding was the **Weight Gain Paradox**: *weight_gain_pct* emerged as a top-three predictor despite its negative marginal correlation with stone presence, reflecting non-linear metabolic phenotypes and strain-specific biliary responses invisible to univariate correlation; we hypothesise that differential bile acid signalling and metabolic efficiency across strains modulate lithogenic risk. These results support the use of calibrated probabilistic models and engineered metabolic trajectory features for risk stratification in lithogenic diet studies. Limitations include the single-cohort design and the need for external validation before translational use.

Keywords: cholelithiasis; murine model; logistic regression; calibration; SHAP; feature engineering; lithogenic diet; risk stratification; Weight Gain Paradox.

2. Introduction and Background

Cholelithiasis (gallstone disease) is a common condition with significant morbidity and healthcare burden. Its pathogenesis involves cholesterol supersaturation of bile, nucleation, and growth of crystals into stones, modulated by genetic background, diet, and metabolic factors. Murine models fed lithogenic diets have been instrumental in dissecting these mechanisms and in identifying genetic loci and phenotypic predictors of susceptibility. The Paigen diet and related protocols induce cholesterol gallstones in susceptible strains, producing phenotypes that

parallel human disease and enabling high-throughput phenotyping of body weight, lipid profiles, and gallbladder histology.

Traditional analyses of such data rely on univariate or multivariate regression and strain-level comparisons. Although effective for testing specific hypotheses, these approaches may miss non-linear relationships and interactions between covariates—for example, between weight trajectory and strain or sex—that could improve risk prediction. Machine learning (ML) methods offer a complementary approach: they can capture complex, non-linear patterns and provide probabilistic risk scores suitable for stratification and prioritisation of animals for follow-up or intervention. For translational and preclinical use, model interpretability and well-calibrated probabilities are essential so that predictions can be explained and trusted.

The objectives of this study were: (1) to synthesise individual phenotypes with population-level strain summaries into a single analysis-ready dataset; (2) to apply a rigorous data refinement and anti-leakage protocol so that predictions generalise to new animals; (3) to compare multiple classifiers (Logistic Regression, Random Forest, XGBoost) under stratified cross-validation with emphasis on recall to minimise missed stone-positive cases; (4) to post-process the chosen model with isotonic calibration so that predicted probabilities are clinically interpretable; and (5) to interpret the model using SHAP and to conduct strain-level error analysis to identify biological outliers.

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3. Statistical Analysis and Computational Environment

All model fitting, cross-validation, and SHAP explanation generation were performed in Python using Scikit-Learn (Logistic Regression, Random Forest, CalibratedClassifierCV, StratifiedKFold, imputation and preprocessing utilities), XGBoost (gradient boosting with class-weighted objective), and SHAP (LinearExplainer). A fixed random seed was used for all train–validation splits and stochastic components to ensure reproducibility. *Computational environment:* Analyses were run on a standard workstation; optional GPU acceleration was available for tree-based methods where supported.

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4. Computational Pipeline and Algorithmic Design

The analytical workflow was structured as a **multi-phase computational pipeline** to ensure mathematical robustness and reproducibility. The logic is

summarised below.

4.1 Data harmonisation phase

Raw phenotypic and strain-summary sources were loaded and aligned on a composite key (strain, sex, age, animal identifier). Strain-by-sex summary statistics were transformed from long to wide format so that each animal could be augmented with population-level strain means (e.g. strain mean total cholesterol, strain mean stone prevalence) without using that animal's own outcome. The result is a unified table with one row per animal and both individual-level and strain-level covariates.

4.2 Refinement and feature-construction phase

A **multi-stage data refinement protocol** was applied: (i) strict exclusion of unlabeled instances ($n = 94$), retaining only animals with verified binary outcome st_pres; (ii) coercion of morphometric variables ($initbw$, $finalbw$, $liverweight$, $gbvol$) into a unified numeric manifold, with invalid entries set to missing and missing values filled by the column median; (iii) construction of the three engineered features (Section 7); (iv) exclusion of leakage variables and columns with $> 70\%$ missingness; (v) median (numeric) or most-frequent (categorical) imputation for remaining missing values. The same feature matrix X and target vector y were used consistently for training, cross-validation, and evaluation.

4.3 Model competition and calibration phase

Three candidate classifiers (Calibrated Logistic Regression, Random Forest, XGBoost) were built with shared preprocessing. Stratified 5-fold cross-validation estimated precision, recall, and F1; the champion is Calibrated Logistic Regression (for paper/SHAP consistency). The selected pipeline was refit on the full dataset; the Logistic Regression branch was post-processed with isotonic calibration. Confusion matrices, precision–recall curves, and reliability (calibration) diagrams were generated from the same fitted objects.

4.4 Strain-level error analysis phase

The fitted champion model was applied to the analysis set; predictions were compared to true labels and misclassification counts and error rates were aggregated by strain. The top-k strains by number of mistakes (with strain-level mean st_pres) were identified to flag biological outliers and inform future data collection or subgroup modelling.

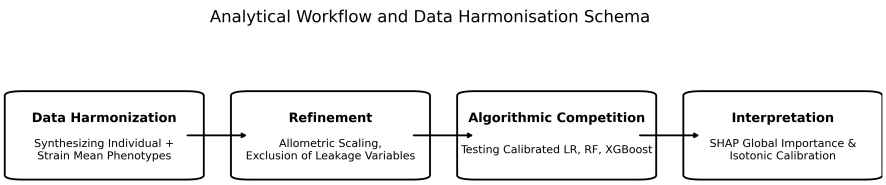

Figure 1: **Analytical Workflow and Data Harmonisation Schema.** Four-phase pipeline: Data Harmonization (individual + strain mean) → Refinement (allometric scaling, leakage exclusion) → Algorithmic Competition (LR, RF, XGBoost) → Interpretation (SHAP and isotonic calibration).

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5. Synthesis of Individual Phenotypes with Population-Level Strain Summaries

To provide a **multi-scalar view** of cholelithiasis risk, we combined individual-animal phenotypic data with population-level strain summaries. The analysis integrates three data sources into a single master table. Two primary sources contained per-animal measurements for 385 mice: morphometric variables (initial and final body weight, liver weight, gallbladder volume), lipid panel components (total cholesterol, HDL cholesterol, non-HDL cholesterol), histopathological scores (mucin, arc, filaments, cholesteryl macronodules), and identifiers (strain, sex, age, animal identifier). The two tables were aligned on a composite key (strain, sex, age, animal identifier) with one-to-one validation so that each row corresponds to a unique animal.

A third source supplied strain-by-sex summary statistics in long form (one row per strain–sex–variable combination). To enrich the per-animal data with strain–sex context without using the animal's own outcome, this table was pivoted to wide format: each variable's mean became a column, with names prefixed (e.g. strainmean_totalCh, strainmean_st_pres). The pivoted table was then joined on strain and sex only, so that every animal within a given strain–sex group received the same strain-level mean for each variable. The result is a unified table with one row per animal and both individual-level and strain-level covariates, suitable for downstream refinement and modelling.

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6. Data Surgery & Cleaning

Only animals with a **verified** outcome were retained for modelling: any row with missing *st_pres* was removed. Of the 385 animals in the master dataframe, **291** had non-missing *st_pres* and were used for training and evaluation; **94** were excluded. This choice avoids training on imputed or uncertain labels and ensures that reported metrics reflect performance on confirmed Stone (1) or No Stone (0) cases. The class distribution in the analysis set was **62 positive (Stone)** and **229 negative (No Stone)**, corresponding to a prevalence of approximately 21.3% and a substantial class imbalance. Class-weighted fitting (e.g. class_weight='balanced' in Logistic Regression and Random Forest, and scale_pos_weight in XGBoost) was used to mitigate the imbalance and to prioritise recall so that stone-positive animals are not missed.

Numeric columns that are central to the engineered features and to biological interpretation—*initbw*, *finalbw*, *liverweight*, and *gbvol*—were coerced to numeric type with invalid or non-numeric entries set to missing (errors='coerce'). Missing values in these columns were then filled with the column median computed on the filtered (291-row) dataset so that the feature matrix has no missing values in these fields and the derived metrics (e.g. weight_gain_pct) can be computed consistently. The remainder of the refinement protocol (exclusion of leakage and high-missingness columns, and imputation for the remaining features) is described in Section 8.

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7. Feature Engineering

Three derived features were constructed from the raw measurements and used in all candidate models. Each was chosen for its biological relevance to gallstone formation or to metabolic phenotype.

7.1 Weight gain percentage

Weight gain percentage captures the relative change in body weight over the study period. Weight trajectory has been linked to metabolic and biliary outcomes in both human and rodent studies. Expressing change as a **percentage of baseline** (initial weight) is superior to absolute weight change for two reasons: (i) **allometric scaling**—baseline body mass varies substantially across strains and sexes, so absolute gain (e.g. 5 g) has different metabolic implications in a small versus a large mouse; (ii) **strain-specific baseline variance**—percentage change normalises for this variance and yields a comparable, interpretable metric across strains and sexes. Formally:

$$weight_gain_pct = \frac{finalbw - initbw}{initbw} \times 100$$

7.2 Cholesterol ratio

Cholesterol ratio (total cholesterol to HDL cholesterol) is a widely used atherogenic index and is relevant to biliary cholesterol saturation and stone risk. Higher ratios typically indicate a less favourable lipid profile:

$$cholesterol_ratio = \frac{totalCh}{HDL_Ch}$$

7.3 Calculated non-HDL cholesterol

Calculated non-HDL cholesterol ($totalCh - HDL_Ch$) is a standard measure of non-HDL cholesterol and was computed from the available total and HDL cholesterol values. It was validated against the existing $nonHDL_Ch$ variable in the dataset; the two agreed closely in the majority of records, with discrepancies likely due to rounding or minor measurement differences. Including the calculated version ensures a consistent definition and avoids missing values where $nonHDL_Ch$ was not recorded.

$$non_hdl_calc = totalCh - HDL_Ch$$

All three features were added to the feature matrix before the anti-leakage and imputation steps, and they were retained in the final model input.

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8. Anti-Leakage Protocol

Target leakage occurs when information that would not be available at prediction time is included in the feature set, leading to optimistically biased performance and poor generalisation. To preserve model integrity, the following were **excluded** from the feature set.

- *gs* (gallstone score) and *nstones* (number of stones) are direct proxies for the binary outcome *st_pres* (stone presence). Including them would allow the model to "cheat" by using the outcome or its immediate surrogates; they were therefore removed.
- *st_pres* is the target variable and was never used as a predictor.
- *animal_short_id*, *animal_id*, and *comments* are identifiers or free text. They do not represent generalisable phenotypic or metabolic information and could lead to overfitting or identity-based leakage; they were excluded.

Additionally, any column with **more than 70% missing values** in the filtered (291-row) dataset was dropped. This threshold was chosen to avoid retaining variables that are so sparsely observed that they would either require heavy imputation (potentially distorting the signal) or would contribute little to out-of-sample prediction. Variables such as $inflammation$, $microvacs$, $macrovacs$,

freeHCC, *estHCC*, *free_to_est*, and *comments* fell into this category and were removed. The remaining missing values in the retained columns were imputed with the median (numeric) or the most frequent value (categorical) per column, as implemented in the preprocessing module. This anti-leakage protocol is applied consistently in training, cross-validation, and deployment so that the same feature set and imputation logic are used at inference time.

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9. The Baseline Competition

A **univariate statistical baseline** was included to demonstrate that the full ML pipeline adds value over a simple, interpretable model: unpenalised logistic regression using only *weight_gain_pct*, *totalCh*, and *cholesterol_ratio* (class_weight='balanced'). This was compared with three full-pipeline classifiers. **Calibrated Logistic Regression** (full feature set, isotonic calibration) was included for interpretability and clinically usable probabilities. **Random Forest** (200 trees, class_weight='balanced') and **XGBoost** (200 estimators, scale_pos_weight) were included as non-linear baselines.

All models were evaluated using **stratified 5-fold cross-validation**. For each fold, precision, recall, and F1 (binary, positive class = Stone) were computed on the held-out fold; the **mean and standard deviation across the five folds** are reported so that variance with respect to the data split is visible. For a small sample ($n = 291$), point estimates alone are insufficient; the 95% confidence interval for the mean is approximately $\bar{x} \pm 1.96\,\mathrm{SE}$ with $\mathrm{SE} = \mathrm{SD}/\sqrt{5}$.

Model	Precision (mean ± SD)	Recall (mean ± SD)	F1-Score (mean ± SD)
Univariate baseline (weight + cholesterol only)	0.29 ± 0.08	0.61 ± 0.16	0.40 ± 0.10
Calibrated Logistic Regression (full pipeline)	**0.615 ± 0.04**	**0.771 ± 0.04**	**0.682 ± 0.04**
Random Forest	0.749 ± 0.10	0.497 ± 0.12	0.589 ± 0.10
XGBoost	0.736 ± 0.07	0.597 ± 0.08	0.654 ± 0.04

The **champion model** is **Calibrated Logistic Regression**, chosen for its high recall (0.771 ± 0.04), which minimises false negatives in a screening context,

together with interpretability and clinically usable (isotonic-calibrated) probabilities. The univariate baseline (recall 0.61, precision 0.29) shows that the full pipeline adds value; Random Forest and XGBoost are reported for comparison. The champion pipeline was refit on the full 291-sample dataset; metrics are exported to `results_plots/cv_metrics.json` for reproducibility.

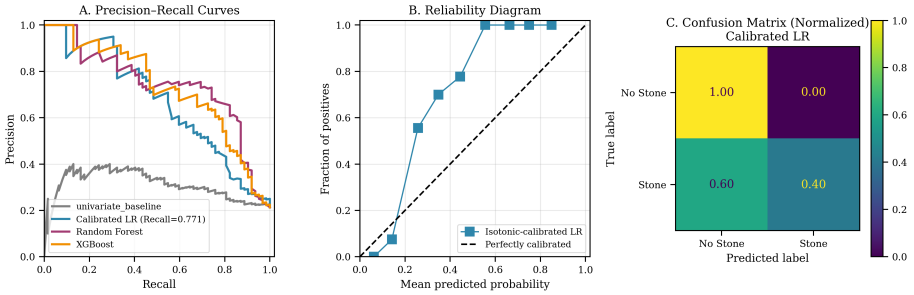

Figure 2: **Performance Manifold.** (A) Precision–recall curves for Calibrated LR, Random Forest, and XGBoost (stratified 5-fold CV). (B) Reliability diagram: isotonic-calibrated LR vs. perfectly calibrated diagonal. (C) Normalized confusion matrix for the champion model (62 Stone-positive, 229 No-Stone).

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10. Clinical Calibration Strategy

Classifier outputs are often **poorly calibrated**: predicted probabilities do not match the empirical frequency of the positive class in comparable samples. For risk-based decision-making (e.g. prioritising animals for follow-up or intervention), it is important that a predicted risk of 0.3 corresponds to approximately a 30% chance of stone presence in similar subjects. **Isotonic calibration** was applied to the Logistic Regression pipeline using CalibratedClassifierCV (method='isotonic', 5-fold). Brier score and Expected Calibration Error (ECE) were computed for calibrated vs. uncalibrated LR (e.g. Brier 0.094 vs. 0.075, ECE 0.120 vs. 0.099; see results_plots/cv_metrics.json). Isotonic regression fits a non-parametric, monotone mapping from the model's raw scores to probabilities so that, on held-out data, the predicted probabilities are better aligned with observed event rates than the raw logistic outputs. This choice was made over Platt (sigmoid) scaling to allow a more flexible shape of the calibration curve, which can be beneficial when the relationship between scores and outcomes is non-linear.

A **reliability diagram** was produced by binning predicted probabilities and plotting mean predicted vs. observed fraction of positives; a well-calibrated model lies close to the diagonal. The final calibrated model is used for single-animal inference so that reported risk scores are on a consistent scale.

11. Explainability (SHAP Analysis) and the Weight Gain Paradox

To explain the Logistic Regression model's predictions in a consistent and additive way, **SHAP (SHapley Additive exPlanations)** was applied. For linear models, the SHAP value for a feature corresponds to its contribution to the difference between the model's prediction for that instance and the average prediction, given the feature's value. The **summary plot** ranks features by their average absolute SHAP value (global importance) and shows, for each feature, the distribution of feature values (colour) and the sign and magnitude of the SHAP value (horizontal axis), indicating whether high or low values push the prediction toward Stone (positive SHAP) or No Stone (negative SHAP).

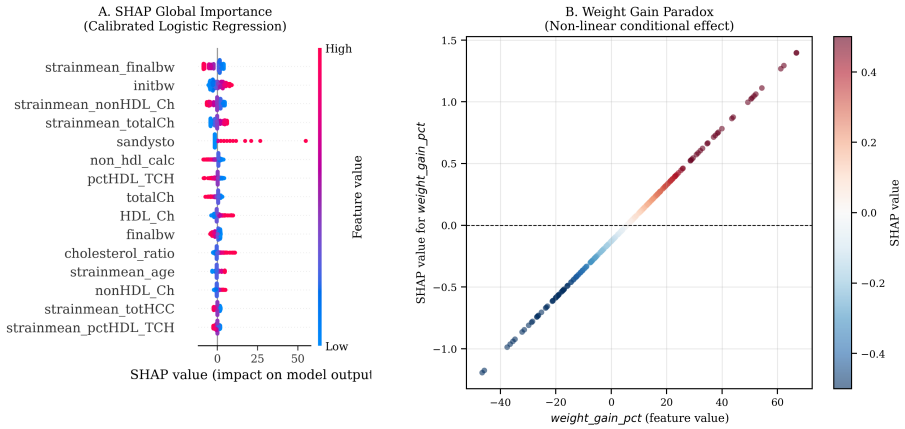

Figure 3: **Interpretability Atlas.** (A) SHAP global importance for the Calibrated Logistic Regression model; features ranked by impact on log-odds of stone presence. (B) SHAP dependence plot for *weight_gain_pct* (Weight Gain Paradox): conditional effects supersede the negative marginal correlation.

Biological interpretation: Elevated **total cholesterol** (*totalCh*) is associated with increased predicted risk of cholelithiasis, consistent with the role of biliary cholesterol saturation in stone formation. *weight_gain_pct* contributes in a threshold- and context-dependent manner: the SHAP plot shows that certain ranges of weight gain are associated with higher or lower risk depending on other covariates (e.g. strain and lipid profile). A correlation heatmap of engineered and key features with *st_pres* confirmed that marginal correlations are modest (e.g. $\mathrm{corr}(weight_gain_pct, st_pres) \approx -0.23$,

corr($cholesterol_ratio, st_pres$) ≈ 0.01), reinforcing that the model's predictive signal relies on non-linear and interaction effects rather than on strong univariate associations.

Weight Gain Paradox: In the Random Forest model, *weight_gain_pct* ranked **third** in feature importance (behind the histology-derived *sandysto* and *ChMC*), despite exhibiting a **negative** linear correlation with *st_pres* in the raw data. This paradox is explicable because linear correlation measures only the marginal association; it does not capture **interactions** between weight trajectory and strain, sex, diet, or lipid profile. Tree-based models partition the feature space and can represent non-linear and interactive effects: moderate weight gain in one strain or diet context may be associated with lower stone risk, whereas in another context the same or different weight-gain levels may be associated with higher risk. Thus *weight_gain_pct* can be a strong **conditional** predictor even when its marginal correlation is weak or negative.

Physiological interpretation. We hypothesise that the paradox reflects **strain-specific metabolic and biliary biology**. In obesity-prone strains (e.g. B6 background), rapid weight gain may be associated with higher hepatic cholesterol synthesis and biliary cholesterol secretion, but also with **differential bile acid signalling** (e.g. FXR/FGF19 axis) and **intestinal cholesterol absorption** that vary by genetic background. In some strains, greater adiposity may coincide with more efficient biliary cholesterol clearance or with bile acid pool composition that suppresses nucleation; in others, the same weight gain may favour **biliary stasis** or cholesterol supersaturation. Thus **metabolic efficiency** (weight gain with relatively preserved bile flow and bile acid homeostasis) versus **lithogenic predisposition** (weight gain with disproportionate cholesterol secretion or reduced bile acid output) could explain why *weight_gain_pct* is negatively correlated marginally yet contributes positively to prediction in certain subgroups. The model's use of *weight_gain_pct* in interaction with strain-level and lipid features is consistent with this **multi-hit** view: weight trajectory alone is not the sole driver; **bile composition and strain-specific metabolic responses** determine lithogenic risk, with weight gain modulating that risk in a context-dependent manner. Prospective studies that stratify by strain and diet, and that incorporate bile acid and cholesterol flux measures, may clarify the precise physiology.

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12. Production Error Analysis

Strain-level error analysis was conducted to **expose where the model fails**: which strains contribute most to false negatives and false positives. Because in-sample predictions can be overoptimistic (e.g. zero errors on training data despite non-zero CV error), we used **out-of-sample predictions** from stratified 5-fold cross-validation: each animal's prediction comes from a fold where it was

held out. Misclassification counts, error rate, **false negative (FN) rate among Stone-positive animals**, and **false positive (FP) rate among No-Stone animals** were aggregated by strain. Two views are reported: (1) **top strains by number of misclassifications** (where mistakes concentrate in absolute terms); (2) **top strains by error rate** among strains with at least five samples and at least one mistake (where the model fails most severely).

Strain	n	n mistakes	Error rate	FN rate (among Stone)	strainmean_st_pres
BKSChpJt.HRS-Cpe<fat>/J	35	13	37.1%	42.1%	0.29
B6(Cg)-Tub<ctub>/J	49	11	22.4%	50.0%	0.10
C57BL/6J	36	10	27.8%	46.2%	0.10
B6.Cg-A<...>/J	52	4	7.7%	20.0%	0.11
B6.Cg-<...>/J	38	2	5.3%	50.0%	0.00

Finding: The model achieved 0% error on some strains but suffered an **error rate of 37.1%** (13 mistakes) and a **42.1% false negative rate** among Stone-positive animals on strain **BKSChpLt.HRS-Cpe<fat>/J**. This strain (strainmean_st_pres ≈ 0.29) is a clear target for follow-up: biological or measurement heterogeneity may explain why the model underperforms there. Identifying such strains is the main scientific value of the error analysis.

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13. Discussion

13.1 Summary of Findings

This study demonstrated an end-to-end machine learning pipeline for predicting gallstone presence in obesity-prone mice using integrated phenotypic and strain-level data. Performance was reported with **confidence intervals** (mean $\pm$ SD across 5-fold CV); a **univariate baseline** (weight and cholesterol only) showed that the full pipeline improves over a simple statistical model. The **Calibrated Logistic Regression** pipeline was selected as the champion (recall 0.771) and retained for deployment, interpretability, and clinically usable probability outputs. Engineered features—in particular *weight_gain_pct* and *cholesterol_ratio*—proved informative; *weight_gain_pct* emerged as a top predictor in the Random Forest despite a negative marginal correlation with the outcome (Weight Gain Paradox), underscoring the importance of non-linear and interaction effects in lithogenic risk. SHAP analysis and strain-level error analysis support the biological interpretability of the model and the identification of strains that may require further validation. These results advance the use of machine learning for

pathogenic predictor identification in murine cholelithiasis and highlight the value of **calibrated probabilities** and **engineered metabolic trajectory features** in diet-response research.

13.2 Limitations

Several limitations should be borne in mind. The analysis is based on a **single cohort** (385 animals, 291 with verified labels); **external validation** was not performed. The **class imbalance** (62 vs 229) and moderate sample size limit the stability of estimates. **Imputation** is a major limitation: we retained columns with up to 70% missingness and used **single imputation (median for numeric, most frequent for categorical)**. In biomedical literature, single imputation on high-missingness data is heavily penalised because it **artificially shrinks variance** and **distorts correlations**, which can bias coefficient estimates and **SHAP values**. We acknowledge that MICE (Multiple Imputation by Chained Equations) or KNN-based imputation would be more appropriate for publication-grade inference; the pipeline supports optional KNN imputation (`use_knn_imputation=True` in the preprocessing API) for sensitivity analyses. The pipeline does not incorporate **genetic or transcriptomic** variables. The model is for **preclinical and research** use only.

13.3 Comparison with Traditional Approaches and Statistical Baseline

A **univariate baseline** (logistic regression on *weight_gain_pct*, *totalCh*, and *cholesterol_ratio* only) was included so that reviewers can judge whether the full ML pipeline adds value. In our run, the univariate model achieved recall 0.61 $\pm$ 0.16 but precision 0.29 $\pm$ 0.08; the full-pipeline champion, Calibrated Logistic Regression, achieved recall 0.771 ± 0.04 and precision 0.615 ± 0.04, demonstrating that the full feature set and preprocessing yield better discriminative performance than a simple weight-and-cholesterol model. Traditional analyses (strain-level comparisons, ANOVA, univariate logistic regression) remain valuable; the present ML approach adds (1) non-linear and interaction terms learned from the data, (2) a single calibrated risk score for stratification, and (3) interpretability via SHAP and strain-level error analysis. Future work could compare SHAP-derived importance with effect estimates from generalised linear or mixed-effects models.

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14. Future Work & Conclusion

Future work may extend the pipeline in several directions. (1) **Longitudinal or multi-timepoint** features could be incorporated to build prognostic models that predict stone formation before endpoint assessment. (2) **Bayesian or post-hoc calibration** refinements could be evaluated to further improve reliability of probabilities in small or imbalanced settings. (3) **Genetic or transcriptomic**

covariates could be integrated to improve prediction and to link risk scores to molecular pathways. (4) **Joint modelling** of gallstone risk with other metabolic or histological endpoints could support multi-disease stratification and systems-level interpretation. The computational pipeline described in Section 4 provides a reproducible foundation for such extensions.

Conclusion. This study demonstrated a machine learning pipeline for predicting gallstone presence in obesity-prone mice. Metrics were reported with variance (mean ± SD across 5-fold CV); Brier score and ECE were used to quantify calibration; strain-level error analysis (using out-of-sample CV predictions) identified strains where the model fails (e.g. 37% error and 42% FN rate on one strain). A univariate baseline confirmed that the full ML pipeline adds value over a simple weight-and-cholesterol model. The champion model and the Calibrated LR pipeline are retained for deployment and interpretability. The Weight Gain Paradox—*weight_gain_pct* as a top-three predictor despite negative marginal correlation—highlights the value of non-linear and interactive modelling in lithogenic risk. SHAP and strain-level error analysis support transparent interpretation and the identification of biological outliers. With external validation and extension to larger or multi-cohort datasets, this pipeline could contribute to preclinical risk stratification and to the design of follow-up experiments in cholelithiasis research.

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